

Result:

Image recognized →

🐼 3D animal appears

Move image →

animal follows.

AR app complete.

---

## Optional Improvements

Add:

- ✓ sound
  - ✓ walking animation
  - ✓ particle effects
  - ✓ multiple images = multiple animals
  - ✓ touch interaction
- 

## Viva Questions

### Q1 What is image recognition AR?

Detect image and place virtual object.

### Q2 What is AR Foundation?

Unity AR framework.

### Q3 What is Reference Image Library?

Stores detectable images.

### Q4 What is Prefab?

Reusable object template.

### Q5 What is tracked image?

Recognized real-world image.

---

Now you have all **15 practicals** covered for your Unity 2022.3 AR/VR lab.



Develop a scene in Unity that includes:

- i. A cube, plane and sphere, apply transformations on the 3 game objects.
- ii. Add a video and audio source

Here is **Practical: Develop a scene in Unity with Cube, Plane, Sphere + apply transformations + add Video and Audio Source**, explained step by step for beginners.

Goal:

Scene includes:

- ✓ Plane
- ✓ Cube
- ✓ Sphere

Apply:

- ✓ Move
- ✓ Rotate
- ✓ Scale

Add:

- ✓ Video playback
- ✓ Audio source/music

This practical covers **objects + transform + multimedia**.

---

## Part 1 — Create Project

Open Unity Hub

Create:

- **3D Core**

Project name:

TransformMediaScene

Click **Create**

---

## Part 2 — Create Plane

Top menu:

**GameObject** → **3D Object** → **Plane**

Rename:

Ground

Transform:

Position:

- X = 0
- Y = 0

- $Z = 0$

Rotation:

- $X = 0$
- $Y = 0$
- $Z = 0$

Scale:

- $X = 1$
- $Y = 1$
- $Z = 1$

Plane created.

---

## Part 3 — Create Cube

Top menu:

**GameObject** → **3D Object** → **Cube**

Rename:

Cube

Set:

Position:

- $X = -2$
- $Y = 1$
- $Z = 0$

Scale:

- $X = 1$
  - $Y = 1$
  - $Z = 1$
- 

## Apply Transformations to Cube

### Move

Change Position:

- $X = -3$

- $Y = 1$
- $Z = 2$

Cube moves.

---

## Rotate

Rotation:

- $X = 45$
- $Y = 45$
- $Z = 0$

Cube rotates.

---

## Scale

Scale:

- $X = 2$
- $Y = 1$
- $Z = 1$

Cube stretches.

Transformation applied.

---

## Part 4 — Create Sphere

GameObject →

**3D Object** → **Sphere**

Rename:

Sphere

Set:

Position:

- $X = 2$
  - $Y = 1$
  - $Z = 0$
- 

## Apply Transformations to Sphere

Move:

- $X = 3$
- $Y = 2$
- $Z = -1$

Rotate:

- $X = 20$
- $Y = 50$
- $Z = 10$

Scale:

- $X = 1.5$
- $Y = 1.5$
- $Z = 1.5$

Done.

---

## Part 5 — Add Materials

Cube:

red

Sphere:

blue

Plane:

green / gray

Scene looks better.

---

## Part 6 — Add Audio Source

Import MP3/WAV file into Assets.

Example:

background music.

Drag file into Project window.

Done.

---

Create Empty object:

Rename:

AudioManager

Inspector →

Add Component →

### Audio Source

Assign imported sound.

Enable:

☒ Play On Awake

☒ Loop

Volume:

0.5

Now sound plays.

---

## Part 7 — Add Video

Import MP4 video into Assets.

Drag into Project window.

Done.

---

Create plane:

GameObject →

3D Object →

Plane

Rename:

Screen

Position:

- X = 0
- Y = 3
- Z = 5

Rotate:

- X = 90

Scale:

2 / 1 / 1

---

## Part 8 — Add Video Player

Select:

**Screen**

Inspector →

Add Component →

Search:

**Video Player**

Add.

Source:

**Video Clip**

Assign imported MP4.

Render Mode:

**Material Override**

Target Renderer:

Mesh Renderer

Material Property:

MainTex

Check:

☒ Play On Awake

☒ Loop

Done.

Video plays on plane.

---

## Part 9 — Lighting

Directional Light:

Intensity:

1.2

Soft shadows ON.

---

## Part 10 — Camera

Position:

- X = 0
- Y = 5
- Z = -10

Rotation:

- $X = 25$

Good view.

## Part 11 — Play

Press 

You will see:

- ✓ transformed cube
- ✓ transformed sphere
- ✓ plane
- ✓ video playing
- ✓ audio playing

Practical complete.

## Viva Questions

### Q1 What are transformations?

Move, Rotate, Scale.

### Q2 What is Transform Component?

Controls object position, rotation, scale.

### Q3 What is Audio Source?

Plays sound.

### Q4 What is Video Player?

Plays video in Unity.

### Q5 What is Play On Awake?

Auto play on start.

This practical is easy and good for viva because it covers **basic objects + transformation + multimedia** together.



Develop a scene in Unity that includes a cube, plane and sphere. Create a new material and texture separately for three Game objects. Change the color, material and texture of each Game object separately in the scene. Write a C# program in visual studio to change the color and material/texture of the game objects dynamically on button click.

Here is **Unity Practical: Cube + Plane + Sphere with separate Material/Texture + change dynamically on button click using C# in Unity**, explained step by step.

Goal:

Scene contains:

- ✓ Cube
- ✓ Plane
- ✓ Sphere

Each has:

- ✓ different color
- ✓ different material
- ✓ different texture

On button click:

- color changes dynamically
- material changes dynamically
- texture changes dynamically

This practical covers:

**Objects + Materials + Textures + UI Button + C# Scripting**

---

## Step 1 — Create New Project

Open Unity Hub

Create:

- **3D Core**

Project name:

MaterialTextureChange

Click **Create**

---

## Step 2 — Create Objects

Create Plane:

**GameObject** → **3D Object** → **Plane**

Rename:

Ground

---

Create Cube:

**GameObject** → **3D Object** → **Cube**

Rename:

Cube

Position:

- X = -3
  - Y = 1
  - Z = 0
- 

Create Sphere:

**GameObject** → **3D Object** → **Sphere**

Rename:

Sphere

Position:

- X = 3
  - Y = 1
  - Z = 0
- 

## Step 3 — Import Textures

Download / use images:

Example:

- wood texture
- metal texture
- marble texture

Drag images into Assets.

Done.

---

## Step 4 — Create 3 Materials

Assets →

Right click →

Create →

Material

Create:

1.

Mat1

2.

Mat2

3.

Mat3

---

## Step 5 — Configure Materials

### Mat1

Color:

Red

Texture:

Wood image

---

### Mat2

Color:

Blue

Texture:

Metal image

---

### Mat3

Color:

Green

Texture:

Marble image

Done.

---

## Step 6 — Assign Materials

Drag:

Mat1 → Cube

Mat2 → Plane

Mat3 → Sphere

Scene becomes colorful.

---

## Step 7 — Create Alternate Materials

Create 3 more:

- Yellow material
- Purple material
- Orange material

Add different textures.

These will be used for button click.

---

## Step 8 — Add UI Button

Hierarchy →

Right click →

UI →

Button

Rename:

ChangeButton

Text:

Change Material

Canvas auto creates.

Done.

---

## Step 9 — Create Script

Project →

Create C# Script:

MaterialChanger

Open in Visual Studio.

Paste:

```
C#
using UnityEngine;

public class MaterialChanger : MonoBehaviour
{
    public Renderer cubeRenderer;
    public Renderer planeRenderer;
    public Renderer sphereRenderer;

    public Material cubeMat1;
    public Material cubeMat2;

    public Material planeMat1;
    public Material planeMat2;

    public Material sphereMat1;
    public Material sphereMat2;

    bool changed = false;

    public void ChangeObjects()
    {
        changed = !changed;

        if (changed)
        {
            cubeRenderer.material = cubeMat2;
            planeRenderer.material = planeMat2;
            sphereRenderer.material = sphereMat2;
        }
        else
        {
            cubeRenderer.material = cubeMat1;
            planeRenderer.material = planeMat1;
            sphereRenderer.material = sphereMat1;
        }
    }
}
```

Save.

Return Unity.

## Step 10 — Create Manager

Hierarchy →

Create Empty

Rename:

GameManager

Attach:

MaterialChanger

## Step 11 — Assign References

Drag into Inspector:

Cube → cubeRenderer  
 Plane → planeRenderer  
 Sphere → sphereRenderer  
 Assign materials:  
 Mat1 / Alt Mat1  
 Mat2 / Alt Mat2  
 Mat3 / Alt Mat3  
 Done.

---

## Step 12 — Connect Button

Select:  
**ChangeButton**  
 Button component →  
 On Click()  
 Click +  
 Drag:  
 GameManager  
 Select:  
**MaterialChanger** → **ChangeObjects()**  
 Done.

---

## Step 13 — Play

Press   
 Click button.  
 Result:  
 Cube → new color/material/texture  
 Plane → new color/material/texture  
 Sphere → new color/material/texture  
 Click again:  
 returns old materials.  
 Dynamic change complete.

---

## Viva Questions

### Q1 What is Material?

Defines surface appearance.

### Q2 What is Texture?

Image mapped on surface.

### Q3 What is Renderer?

Displays object.

### Q4 What is Button OnClick()?

Calls function on button press.

### Q5 Why use public Material?

Assign from Inspector.

This is a very good practical because it combines **3D objects + material + texture + UI + scripting** in one scene.



Develop and deploy a simple marker based AR app in which you have to write a C# program to play video on tracking a particular marker.

Here is **Unity AR Practical: Marker-Based AR App that plays a video when a particular marker is tracked** using **Unity + C#**, explained step by step for beginners.

Goal:

Phone camera opens 

When a marker/image is detected:

 Marker recognized

→  Video starts playing on/above marker

When marker disappears:

→  Video stops

This is a very common AR practical.

We will use:

- ☒ AR Foundation
- ☒ ARCore
- ☒ Image Tracking
- ☒ Video Player
- ☒ C# Script

## Step 1 — Create Project

Open Unity Hub

Create:

- **3D Core**

Project name:

ARVideoMarker

Click **Create**

---

## Step 2 — Install Packages

Window → Package Manager

Install:

- ☒ AR Foundation
- ☒ ARCore XR Plugin

Done.

---

## Step 3 — Enable XR

Edit → Project Settings → XR Plug-in Management

Enable:

- ☒ ARCore

Done.

---

## Step 4 — Switch Platform

File → Build Settings

Select:

Android

Click:

Switch Platform

Done.

---

## Step 5 — Create AR Session

Hierarchy →

XR →

**AR Session**

Create.

Create:

**XR Origin (AR)**

Done.

---

## Step 6 — Add Image Tracking

Select:

**XR Origin**

Add Component:

**AR Tracked Image Manager**

Done.

---

## Step 7 — Create Marker Image

Choose any image/logo.

Example:

- company logo
- photo
- symbol

Save JPG/PNG.

Import into Assets.

Inspector:

Texture Type:

**Sprite (2D and UI)**

Done.

---

## Step 8 — Create Reference Image Library

Assets →

Create →

XR →

Reference Image Library

Name:

VideoLibrary

Add marker image.

Set physical width:

**0.15**

Assign to tracked manager.

Done.

---

## Step 9 — Import Video

Drag MP4 file into Assets.

Example:

birthday clip / ad / intro video.

Done.

---

## Step 10 — Create Video Screen

GameObject →

3D Object →

Plane

Rename:

VideoScreen

Scale smaller:

0.2–0.3

Rotate so visible above marker.

Position:

Y = 0.1

Done.

---

## Step 11 — Add Video Player

Select:

**VideoScreen**

Add Component:

**Video Player**

Assign:

MP4 clip.

Set:

Source:

**Video Clip**

Render Mode:

**Material Override**

Target:

Mesh Renderer

Material property:

**\_MainTex**

Uncheck:

Play On Awake

Done.

---

## Step 12 — Make Prefab

Drag:

**VideoScreen**

into Assets.

Prefab created.

Delete scene copy.

Done.

---

## Step 13 — Create Script

Create C# Script:

ARVideoController

Paste:

```
C#  
using UnityEngine;  
using UnityEngine.Video;  
  
public class ARVideoController : MonoBehaviour  
{  
    VideoPlayer vp;  
  
    void Start()  
    {  
        vp = GetComponent<VideoPlayer>();  
    }  
  
    void OnEnable()  
    {  
        vp.Play();  
    }  
}
```

```

    }
    void OnDisable()
    {
        vp.Stop();
    }
}

```

Save.

Attach script to:

**VideoScreen prefab**

Done.

---

## Step 14 — Assign Prefab

Tracked Image Manager →

Tracked Image Prefab →

assign:

**VideoScreen prefab**

Done.

---

## Step 15 — Build APK

Build And Run.

Install on Android phone.

Grant camera permission.

---

## Step 16 — Test

Open app.

Point camera at marker.

Result:

Marker detected →

 video plays

Move marker →

video follows marker

Hide marker →

 video stops

Practical complete.

---

# Viva Questions

## Q1 What is marker-based AR?

Uses image marker recognition.

## Q2 What is AR Tracked Image Manager?

Tracks images in camera.

## Q3 What is Video Player component?

Plays video.

## Q4 Why OnEnable()?

Play when shown.

## Q5 Why OnDisable()?

Stop when hidden.

This practical is impressive because it combines **AR + tracking + video playback + scripting** in one project.



Develop and deploy an AR app, implement the following using Vuforia Engine developer portal:

- i. Plane detection
- ii. Marker based Tracking(Create a database of objects to be tracked in Vuforia)
- iii. Object Tracking

Here is **Unity AR Practical using Vuforia Engine + Unity** for:

1. **Plane Detection**
2. **Marker Based Tracking (create database in Vuforia)**
3. **Object Tracking**

This is a big practical, so I'll explain **each step carefully**.

## What is Vuforia?

Vuforia Engine is an AR SDK used in Unity for:

- ✓ Image tracking
- ✓ Object tracking
- ✓ Ground/plane detection
- ✓ Model targets

It is easier for college practicals than coding everything manually.

## Part A — Create Vuforia Account

Open official website:

[Vuforia Engine](#)

Click:

**Register**

Create free account.

Login.

---

## Part B — Get License Key

Dashboard →

**License Manager**

Click:

**Get Development Key**

Name:

MyARProject

Check agreement.

Create.

Copy license key.

Save it.

Needed later.

---

## Part C — Create Unity Project

Open Unity Hub

Create:

**3D Core**

Project name:

VuforiaAR

Create project.

---

## Part D — Install Vuforia in Unity

Open:

## **Window → Package Manager**

Install package:

### **Vuforia Engine AR**

(If package not visible, add package by name/package source.)

After install:

Go:

### **Edit → Project Settings → XR Plug-in Management**

Enable Vuforia if shown.

Done.

---

## **Part E — Enable Vuforia**

Go:

### **Edit → Project Settings → Player**

Find:

**Vuforia Augmented Reality**

Enable.

Restart if needed.

---

## **Part F — Add AR Camera**

Delete:

**Main Camera**

Hierarchy →

GameObject →

Vuforia Engine →

**AR Camera**

Create.

Inspector:

Paste License Key.

Done.

---

=====

## **i) PLANE DETECTION**

=====

Plane detection = detect flat surfaces like:

table / floor / desk

and place object.

## Step 1 — Add Plane Finder

Hierarchy →

GameObject →

Vuforia Engine →

**Plane Finder**

Create.

Also creates:

**Ground Plane Stage**

Done.

## Step 2 — Create Object

Create cube/sphere/character.

Rename:

PlacedObject

Scale smaller.

Drag into Assets →

make prefab.

Delete scene copy.

## Step 3 — Assign Prefab

Select:

**Plane Finder**

Inspector →

Content Positioning Behaviour →

assign prefab.

Done.

---

## Step 4 — Run

Build APK.

Open on phone.

Scan floor/table.

Tap screen.

Object appears on detected plane.

Plane detection complete.

---

=====

## ii) MARKER BASED TRACKING

=====

Detect marker image and show object.

---

## Step 1 — Create Database

Open:

[Vuforia Target Manager](#)

Click:

**Add Database**

Name:

MyMarkers

Type:

Device.

Create.

---

## Step 2 — Add Marker

Inside database:

**Add Target**

Choose:

Image Target

Upload image:

logo/photo/card.

Width:

10

Name:

Marker1

Process image.

Done.

---

## Step 3 — Download Database

Download for:

Unity Editor.

Import package into Unity.

Done.

---

## Step 4 — Add Image Target

Hierarchy →

GameObject →

Vuforia Engine →

**Image Target**

Create.

Database:

select:

**MyMarkers**

Target:

**Marker1**

Done.

---

## Step 5 — Add 3D Object

Create:

Cube / Car / Animal.

Make child of:

ImageTarget.

Set position:

Y=0.1

Done.

---

## Step 6 — Test

Point camera at marker.

Object appears.

Marker tracking complete.

---

=====

## iii) OBJECT TRACKING

=====

Recognize real object:

bottle / toy / box

and track it.

---

## Step 1 — Prepare Object Scan

Need real object with rich texture.

Examples:

✓ toy car

✓ bottle label

✓ box

Good tracking surface.

---

## Step 2 — Create Object Target

In Vuforia portal:

Add Target →

Object Target

Upload scanned model/object data.

Create target.

Done.

---

## Step 3 — Download Database

Download Unity package.

Import into Unity.

---

## Step 4 — Add Object Target

Hierarchy →

GameObject →

Vuforia Engine →

**Object Target**

Choose database.

Select object target.

Done.

---

## Step 5 — Add AR Content

Child object:

animal / text / 3D model.

Done.

---

## Step 6 — Test

Camera detects real object.

Overlay appears.

Object tracking complete.

---

## Deployment

File →

Build Settings →

Android →

Build And Run.

Install APK.

Grant camera permission.

Done.

---

## Viva Questions

### Q1 What is Vuforia?

AR SDK.

### Q2 What is plane detection?

Detect flat surfaces.

### Q3 What is marker tracking?

Detect predefined image.

### Q4 What is object tracking?

Track real 3D object.

### Q5 What is database in Vuforia?

Stores targets to detect.

---

This one practical covers **3 major AR concepts**, so it is very important for viva and practical exams.